Study-Mate – Chat History

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User:

Insertion sort

Assistant:

Algorithm for Insertion Sort:

```
Algorithm InsertionS ort (a, n)
{
 // sort the array a[1:n] into non decreasing order.
  for i := 2 to n do
 {
  key=a[i];
      // Insert a[i] into the sorted part of the array, i.e., a [1: i -1]
      j=i-1;
  while (j>0 and a[j]>key) do
  {
   a[j+1]=a[j];  // move a[j] to its next position in the right side
   j=j-1;
  }
  a[j+1]=key;

}
}
```

Performance Analysis:
Insertion sort takes linear time in the best case and quadratic time in the worst case. The complexity of insertion sort is O(n^2) because O(n^2) covers linear time as well.